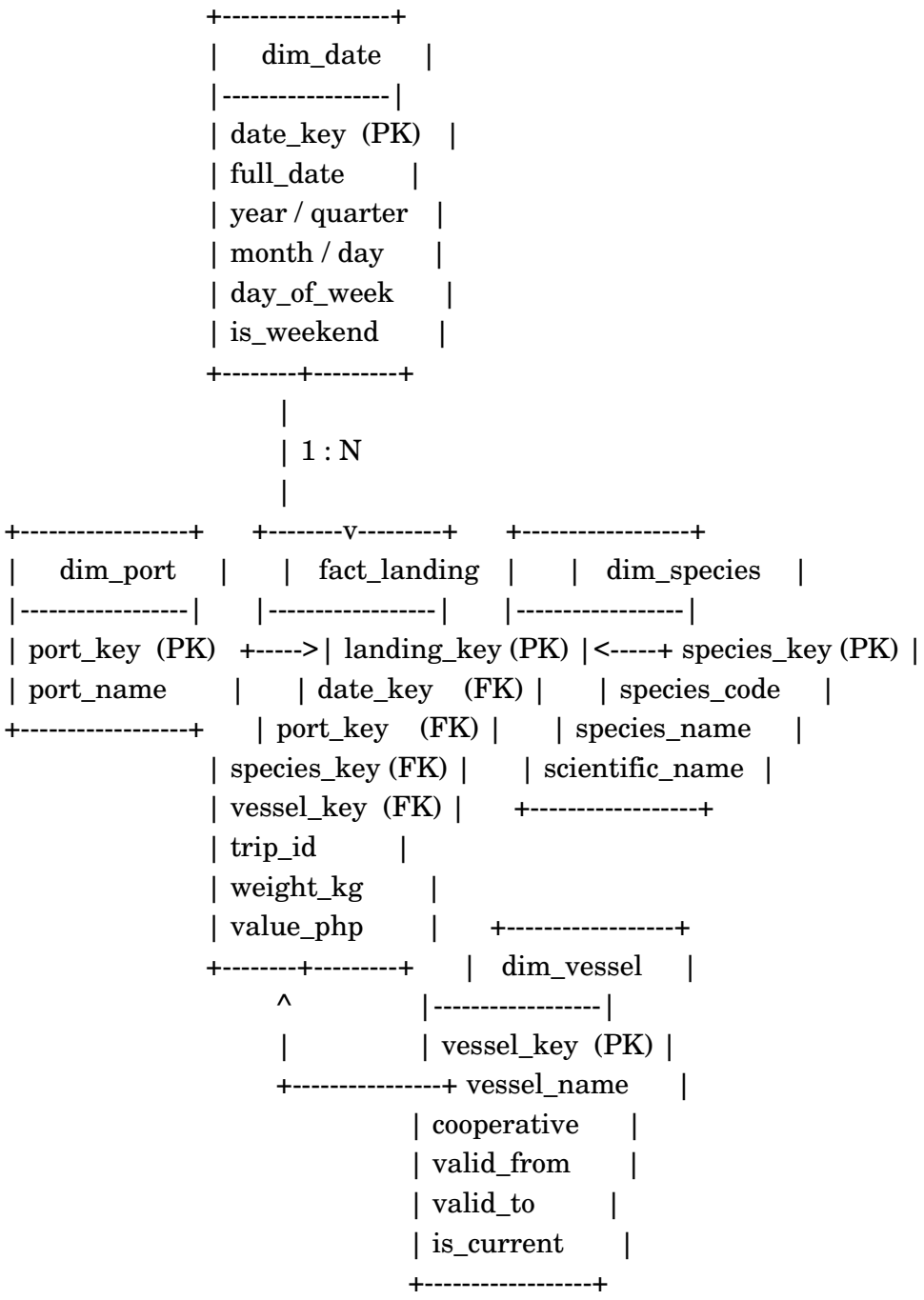
Data Quality Report — Fisheries Landings ETL

Author: Mai Kie **Date:** 2026-11-03 **Source:** messy_landings_2026.csv (20 rows)
Target: dw.fact_landing

1. Star Schema Diagram



Grain of fact_landing: one row per landed trip.

2. Issues Found in Source CSV

#	Issue	Row(s)
1	Missing landing_date	T007
2	Missing weight_kg	T009
3	Negative weight_kg / value_php	T010
4	Extreme outlier weight_kg = 99999	T011
5	Exact duplicate row	T006 (matches T005)
6	Inconsistent species naming (YFT vs Yellowfin Tuna, SJK vs skipjack tuna)	Multiple

3. Rules Applied

- **Null-guard**: reject rows with blank landing_date or blank weight_kg.
- **Domain check**: reject weight_kg <= 0.
- **Outlier rule**: reject weight_kg > 10000 kg (99999 kg is implausible for one trip).
- **Deduplication**: ROW_NUMBER() OVER (PARTITION BY all business columns EXCEPT trip_id) — keep rn = 1.
- **Standardization**: LOWER(TRIM(species_name)) joined to stg.species_mapping.
- **SCD Type 2 lookup**: join dim_vessel on landing_date BETWEEN valid_from AND valid_to.

4. Rows Rejected

Reason	Count
Missing date	1
Missing weight	1
Negative weight	1
Outlier (> 10000 kg)	1
Duplicate	1
Total rejected	**5**
Rows loaded into fact_landing	**15**